

Detailed Summary of the Paper

“GR00T N1: An Open Foundation Model for Generalist Humanoid Robots”

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Abstract

The paper presents GR00T N1, an open foundation model for generalist humanoid robots. It aims to provide a robotic “mind” capable of vision-language reasoning and generating fluid motor actions across multiple robotic platforms. The model belongs to the Vision-Language-Action (VLA) class with a “dual-system” architecture. It is trained on a wide variety of data (human videos, simulations, real trajectories) organized according to a “data pyramid”. The results show that GR00T N1 outperforms imitation learning methods across several simulation benchmarks and demonstrates effectiveness on real robots.

1. Introduction

Generalist robots must combine three ingredients: (1) versatile hardware, (2) general models capable of tackling multiple tasks, and (3) an effective data strategy. This work draws inspiration from AI “foundation models” and aims to apply this paradigm to robotics, overcoming the problem of “data-islands” (execution examples or robots that are too different from each other). Therefore, GR00T N1 is proposed: a VLA model with a dual-system architecture that supports different embodiments and is trained end-to-end.

2. The GR00T N1 Model

2.1 Model Architecture

The model is structured into two interconnected systems:

- **System 2 (Vision-Language):** processes visual observations and linguistic instructions. It uses a pre-trained Vision-Language Model (VLM) (Eagle-2) to encode images and text into shared tokens. (It works at 10Hz, because it is used to understand the purpose of the action and the goal, so it does not need to run too frequently).
- **System 1 (Action):** a Diffusion Transformer (DiT) that generates high-frequency motor actions conditioned by the tokens from module 2 and the robot’s state. (It works at 120Hz, because it must very frequently tell the robot where to go and what movements to make).

To handle different embodiments, the model includes embodiment-specific MLP encoders that project state and action into a common embedding space. Given a robotic state s_t , observations o_t , and instruction L , the Vision-Language module produces the embedding $z = \text{VLM}(o_t, L)$ (this means that since robots can be very different, the model does not directly predict actions but predicts a vector space common to all and non-specialized). The state is encoded as $e_s = \text{MLP}_s(s_t)$. The action is generated by the DiT, which takes a noisy input and the condition (z, e_s) , learning the denoising field via flow matching (therefore, from the general

and noisy embedding, it extracts the action by learning how to perform denoising). The training loss follows the flow-matching formula (eq. 1 of the paper).

2.2 Data Generation and the Data Pyramid

The data is organized according to a pyramid (from the base to the top, from the most frequent to the least frequent data):

- **Base:** egocentric and web-scale human videos, without explicit actions.
- **Middle layer:** robotic simulations and synthetic trajectories.
- **Top:** real data from physical robots (teleoperation).

To leverage actionless videos:

- **Latent-action codebook** (VQ-VAE) to represent latent actions.
- **Inverse Dynamics Model** (IDM) to generate pseudo-actions.

2.3 Training

Training is split into two phases: pre-training (a mix of data from the pyramid) and post-training (adaptation to specific tasks). Even with few examples, the pre-trained model maintains good performance.

3. Experimental Results

3.1 Simulation

Evaluated on:

- RoboCasa (24 tasks)
- DexMimicGen (9 bimanual tasks)
- GR-1 Humanoid (24 tasks)

Method	RoboCasa	DexMimicGen	GR-1
BC Transformer	26.3%	53.9%	16.1%
Diffusion Policy	25.6%	56.1%	32.7%
GR00T-N1-2B	32.1%	66.5%	50.0%

3.2 Real Robot

On a real GR-1 humanoid: average success rate of 76.8%. With 10% of the data: GR00T N1 achieves 42.6% vs 10.2

3.3 Limitations

Actions are limited to short-horizon tabletop tasks; it does not include complex locomotion.

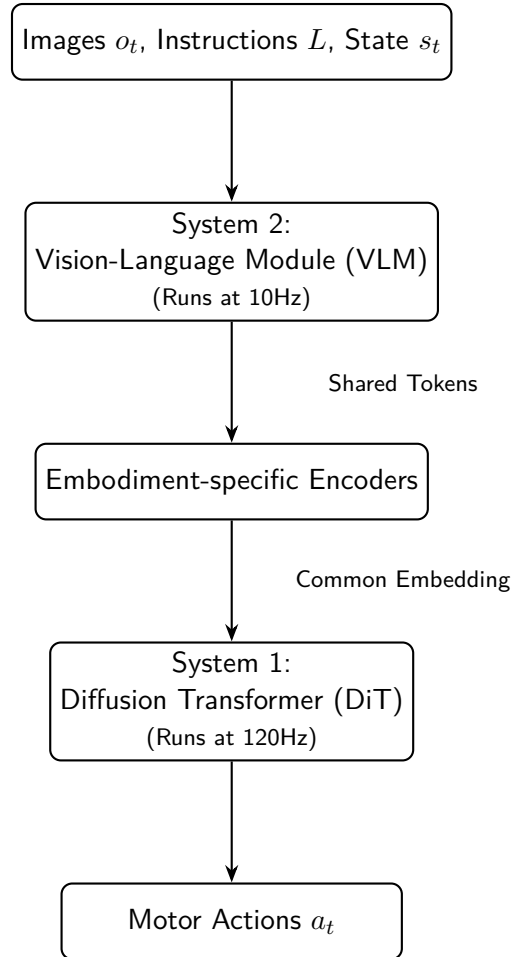
4. Main Contributions

- First open foundation model for generalist humanoids.
- Dual-system architecture (Vision-Language + Diffusion Transformer).
- Data pyramid with human, synthetic, and real sources.
- Multi-embodiment support.
- Few-shot robustness on real robots.

5. Conclusions

GR00T N1 demonstrates that foundation models can be effectively applied to robotics. Future developments: locomotion, large environments, and better linguistic-spatial integration.

6. Structure Diagram of GR00T N1



The diagram illustrates the main pipeline: vision/language $\rightarrow$ encoding $\rightarrow$ action generation.